

**ATTACHMENT 10 – RESPONSE LETTER FROM BECHTEL / PARSONS
BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN
CONTINENTAL COMPANY DATED OCTOBER 12, 1999**
(3 pages)



Massachusetts Turnpike Authority
Central Artery/Tunnel Project

October 12, 1999

Letter No.: C09B2-0350
File No.: CN-9.09B2.02
CN-9.09B2.39.24

Mr. Robert Coutts
Project Manager
Modern Continental Companies, Inc.
600 Memorial Drive
Cambridge, MA 02210

Subject: Central Artery (I-93)/Tunnel (I-90) Project
Contract No. 99119-C09B2
I-90 Seaport Access Tunnel Finishes
Adhesive Anchor Movement

Reference: MCC Letter No. 0114, dated 10/07/99

Dear Mr. Coutts:

As discussed during our Engineering Issues meeting of October, 8 1999, and in response to the above referenced letter, the following was discussed.

We concur that the adhesive anchors were load tested in accordance with the design requirements, however we cannot confirm that the adhesive anchors are correctly loaded. A review of the ceiling system indicates that several turnbuckles are loose. This could induce excessive loading to the point that the anchors may be over stressed which may cause movement.

Other possibilities exist which could cause anchor bolt failure. During erection of the ceiling module the lifting system may have been released in such a manner to cause excessive stress to the anchors (too quickly) or prior to uniform loading of all turnbuckles. Failure to properly prepare & clean the drilled holes could cause a bond failure of the adhesive to the concrete over time.

The following action items were discussed and agreed to:

1. MCC to contact the supplier of the adhesive anchor bolt system, review the installation of the failed anchors and obtain a manufacturer's recommendation.
2. MCC to confirm that all ceiling hangers are correctly loaded and have uniform load.
3. MCC to investigate with the manufacture the possibility to retest the anchors to design loading after the supports are properly adjusted.



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4. B/PB will confirm that ceiling load calculations are correct.

In summary, MCC shall confirm that the ceiling system is properly installed and all connection points are providing the correct support. Anchors that have moved since ceiling installation shall be abandoned and/or removed and replaced. When replacing the defective anchors, insure that proper measures are provided to support the ceiling so that other over stressed conditions are not created.

If you have any questions, please contact me at 350-7594, ext. 56.

Sincerely,

BECHTEL/PARSONS BRINCKERHOFF



Robert E. Steffy
Resident Engineer
Authorized Representative of
the Project Director

RES/ha

cc: M. Charney